Vishal Kumar Rai

Ph.D. Candidate, Economics
Economics and Planning Unit (EPU)
Indian Statistical Institute – Delhi
Fields: Structural Macroeconomics • Life-Cycle Modeling • Ageing and Retirement

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EDUCATION

Ph.D. Candidate in Economics Expected 2026
Indian Statistical Institute – Delhi
Pre-submission seminar completed

M.A. in Economics 2019
Delhi School of Economics

B.A. (Hons.) Business Economics 2015
University of Delhi

DISSERTATION RESEARCH

Essays on Life-Cycle Decisions under Environmental and Epidemiological Shocks

Supervisor: Prof. Monisankar Bishnu

Chapter 1: Climate, Pollution and Endogenous Retirement
Examines how environmental risk shapes life-cycle decisions within a framework jointly determining consumption, health expenditure, environmental abatement, and retirement decisions. The analysis characterizes the mechanisms linking pollution, health investment, and labor supply behavior, demonstrating that environmental conditions can fundamentally alter the relationship between health expenditure and retirement timing.

Chapter 2: Endogenous Retirement in the Presence of a Pandemic
Studies life-cycle behavior under pandemic risk in a model with endogenous consumption, health expenditure, and retirement decisions. Analytical and quantitative results show that pandemic shocks generate heterogeneous responses across age cohorts, affecting health investment and labor supply decisions through mortality, labor-market, and disease-risk channels.

Chapter 3: Occupational Variation in Work Strain: A Life-Cycle Model of Retirement
Introduces occupational heterogeneity into a life-cycle framework by allowing work strain to vary across occupations. The analysis demonstrates how differences in working conditions interact with health expenditure and survival dynamics to generate systematic variation in labor supply and retirement behavior.

RESEARCH PROJECTS

Health, Endogenous Retirement and Demand for Annuity Ongoing
Department of Science and Technology (DST), Government of India
Principal Investigator: Prof. Monisankar Bishnu

Contributed to the development, calibration, and numerical implementation of a lifecycle model of health, ageing, retirement, and annuity demand. Constructed occupation-specific wage, mortality, survival, and frailty profiles using PLFS, LASI, SRS, CDC, and NOMS data; estimated model parameters, implemented numerical optimization in Julia, and conducted policy simulations under alternative pension and tax regimes.

CONFERENCE PRESENTATIONS

Endogenous Retirement in the Presence of a Pandemic
Presented at the Macroeconomics Session, CDE–IEDS International Conference on *Economic Development, Demographic Transition and Environmental Sustainability*, Delhi School of Economics, New Delhi, June 2026.

Accepted from 563 submissions; 113 papers presented from 15 countries.

COMPUTATIONAL TOOLS

SwitchingControl.jl
Registered Julia package for solving endogenous regime-switching optimal control and dynamic economic models using event-driven differential equation methods.

SKILLS

Programming: Julia
Methods: Structural Modeling, Calibration, Dynamic Programming, Numerical Methods
Data: LASI, NFHS, SRS, MEPS, CPS, CDC Life Tables
Tools: LaTeX, Git

TEACHING EXPERIENCE

Teaching Assistant, Indian Statistical Institute – Delhi
Prof. Chetan Ghate

Macroeconomics I and Global Macroeconomics for M.S. Quantitative Economics and Ph.D. students; conducted tutorials and assisted with assessment and course coordination.

REFERENCES

Monisankar Bishnu (Supervisor)
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